

Toheeb Goodluck Ogunade

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Technical Skills

- Languages: Python, Go (Golang), SQL, TypeScript
- ML Frameworks: PyTorch, TensorFlow, Scikit-learn, spaCy, coreferee, GLiNER
- Tools & Infra: FastAPI, Docker, Git, Linux, RunPod
- Competencies: LLM Evaluation, Empirical ML Research, Agent Architecture, Symbolic Reasoning, NLP Pipelines, RAG Systems

Experience

AI/ML Engineer (Contract) Jul 2025 – Jan 2026

OysterSkin

- Engineered and deployed production ML models for personalized recommendation, reducing model retraining cycle times by 30% through automated feature extraction pipeline design.
- Built low-latency inference infrastructure integrating ML models into high-availability web services via Python and FastAPI.
- Conducted A/B testing and statistical model evaluation to optimize predictive accuracy and drive user retention metrics.

AI Intern

Apr 2025 – May 2025

Curacel

- Engineered ML modules for automated health insurance claims processing and high-precision fraud detection; won **Best AI Innovation** at the Curacel Hackathon (2024) for a multilingual healthcare AI assistant with NLP-driven anomaly detection.
- Contributed to scalable AI infrastructure design supporting high-volume, real-time health data streams.

Projects

Velarix — AI Safety & Interpretability Engine | Go, Python, TypeScript <https://github.com/qeinstein/velarix>

- Designed a symbolic mechanistic interpretability layer for AI agents: 100% verifiable causal traces of agent reasoning, non-monotonic belief revision, and counterfactual auditing for root-cause analysis of agent behavior.
- Eliminates logical hallucinations by offloading stateful reasoning to a deterministic Go kernel, bypassing probabilistic LLM errors and attention dilution in long-context settings.
- Shipped LangGraph, CrewAI, LlamaIndex, and OpenAI integration surfaces; fact-extraction pipeline (Delta) achieves p50 35.6 ms / p99 56.3 ms and 96.3% triple recall at zero LLM inference cost.

TurboQuant — LLM KV Cache Compression | Python

<https://github.com/qeinstein/TurboQuant>

- Implemented TurboQuant (arXiv:2504.19874): a two-stage KV cache quantization pipeline combining Lloyd-Max scalar quantization with a 1-bit QJL inner product corrector, achieving $4.7\times$ compression at 4-bit keys + 2-bit values with cosine similarity ~ 0.92 vs fp16.
- Ran 82-test benchmark suite (unbiasedness, MSE bounds, bit-packing, end-to-end correctness) on WikiText-2/GPT-2; baseline perplexity gap: +21.11 over fp16.
- Identified and empirically confirmed an unexplored finding: doubling the QJL sketch size to $m = 2d$ — not tested in TurboQuant, QJL, or PolarQuant — cuts the perplexity gap from +21.11 to +9.83, less than half the quality loss at modest compute overhead.

Research

Context Collapse in LLMs: Semantic Interference vs. Context Length

Published — Medium, Mar 2026

github.com/qeinstein/context_collapse | medium.com/@ogunadetoheeb4

- Ran 560 controlled trials across arithmetic, retrieval, logic, and instruction-following tasks; random context up to $\sim 4,000$ words reduced accuracy by only 1.3% ($98.8\% \rightarrow 97.5\%$), while 1,000 words of semantic noise drove accuracy to 82.5%.
- Identified distractor adoption as the mechanism of failure: 24% of retrieval errors under semantic noise were values sourced directly from irrelevant context, with 100% instruction compliance maintained throughout — isolating answer selection as the failure mode.

A Matched Spectral Benchmark of Quantum Inspired Feature Maps

arXiv:2605.24324 — May 2026

arxiv.org/abs/2605.24324 | with Taofeek Kassim, Etinosa Osaro

- Designed and executed a 10-dataset, 110+-run empirical benchmark comparing amplitude, angle, and basis QIE encodings against 5 matched classical baselines under controlled budgets, using PyTorch and TensorFlow.
- Ran 30 paired statistical comparisons (t-test + Wilcoxon + Cohen's d): 27/30 significantly worse, 0 wins; attributed failures via spectral diagnostics — amplitude rank collapse (erank $\downarrow 1.04$, $\kappa \uparrow 5.7 \times 10^9$), angle geometric equivalence to linear ($\text{CKA} \geq 0.95$ on 7/10 datasets), basis Hamming geometry mismatch.
- Confirmed a falsifiable boundary condition: when $\kappa \approx 2$ and effective rank is high, amplitude encoding achieves near-parity with the best classical baseline ($d = -0.03$, $p = 0.96$).

Education

University of Lagos

B.Sc. Computer Science

Lagos, Nigeria

Expected Jan 2028

Awards & Recognition

Winner, Best AI Innovation — Curacel Hackathon (2024)

8th Place — National Mathematics Olympiad (2023)